Social Media Engagement Analytics using Python

1. Project Overview and Objective

1.1 Project Overview

This project analyzes a social media engagement dataset containing 5000 records across multiple platforms and countries. The analysis covers data cleaning, transformation, exploratory analysis, statistical analysis, and visualization to identify meaningful patterns in user behavior and content performance.

The workflow uses Pandas, NumPy, Matplotlib, Seaborn, and Plotly for data preparation, analysis, visualization, and insight generation.

1.2 Project Objective

The main objective is to transform raw social media engagement data into structured analytical results and meaningful insights about content performance, user behavior, device usage, posting time, and sentiment.

2. Data Sources

Item	Details
Dataset	social_media_engagement_5000.csv
Domain	Social Media / User Engagement
Records	5000 rows × 19 columns
Tool Used	Google Colab
Programming Language	Python 3
Libraries	Pandas, NumPy, Matplotlib, Seaborn, Plotly

3. Project Objectives

- Load and clean the social media engagement dataset.
- Handle missing values and duplicate records.
- Perform exploratory data analysis.
- Create new engagement and transformed metrics.
- Generate statistical insights.
- Build 13 visualizations using Matplotlib, Seaborn, and Plotly.
- Extract actionable insights about content performance.

4. Attribute (Column / Feature) Details

Attribute Name	Description
user_id	Unique user identifier
age	User age
gender	User gender
country	User country
post_type	Type of post: image, reel, text, or video
post_category	Category of post
likes	Number of likes
comments	Number of comments
shares	Number of shares
watch_time_sec	Watch time in seconds
impression_count	Total impressions
posted_at	Post date and time
follower_count	Number of followers
is_verified	Verified account status

device_type	Device used: mobile, tablet, or desktop
sentiment	Post sentiment
hashtags	Hashtags used
engagement_rate	Engagement rate

5. Tools & Technologies

Google Colab was used as the development and execution environment. Python was used for data loading, cleaning, transformation, statistical analysis, visualization, and insight generation.

- Python 3
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly

6. Data Setup & Processing

6.1 Data Import & Setup

- Imported the CSV dataset using Pandas.
- Verified the dataset structure and data types.
- Converted posted_at to datetime format.

6.2 Data Cleaning

- Detected missing values in six columns.
- Filled numerical columns with median values.
- Filled categorical columns with mode values.
- Removed duplicates.
- Standardized gender and sentiment labels.
- Fixed unrealistic values.
- Extracted hashtag count.

6.3 Data Exploration

- Reviewed head, tail, shape, info(), dtypes, and describe().
- Analyzed categorical distributions.
- Created a correlation matrix.
- Grouped data by post_type and country.

6.4 Data Wrangling

- Created an engagement_score metric.
- Added log-transformed metrics.
- Extracted hour, day, and month from posted_at.
- Performed groupby summaries.
- Combined verified and non-verified DataFrames.

7. Python Analysis Structure

7.1 Exploratory Data Analysis

The dataset structure, data types, summary statistics, categorical distributions, correlations, and grouped summaries were examined to understand social media engagement patterns. ***7.2 Feature Engineering***

New features included engagement_score, log-transformed metrics, hashtag count, and time-based fields such as hour, day, and month.

7.3 Statistical Analysis

Descriptive statistics included mean, median, mode, standard deviation, variance, percentiles, skewness, and kurtosis.

7.4 Visualization Analysis

Matplotlib, Seaborn, and Plotly were used to visualize relationships, distributions, category comparisons, correlations, and engagement patterns.

8. Core Analytical Tasks

Task	Description	Python Concepts / Tools
T1	Data Import & Setup	Pandas, datetime conversion, inspection

T2	Data Cleaning	Missing values, duplicates, standardization
T3	Data Exploration	Head/tail, shape, info, describe, groupby
T4	Data Wrangling	Feature engineering, transformations, grouping
T5	Statistical Analysis	Mean, median, mode, variance, percentiles, skewness, kurtosis
T6	Visualization	Matplotlib, Seaborn, Plotly
T7	Final Insights	Content, user, device/time, and sentiment analysis

9. Visualization Analysis

Plot	Chart Type	Analysis
Plot 1	Scatter	Likes vs Impressions
Plot 2	Line	Daily Engagement Trend
Plot 3	Bar	Posts by Category
Plot 4	Pie	Gender Distribution
Plot 5	Histogram	Age Distribution
Plot 6	Box	Engagement Rate
Plot 7	Count Plot	Post Type
Plot 8	Bar	Average Likes by Category
Plot 9	Violin	Followers vs Sentiment
Plot 10	Pair Plot	Numeric Features
Plot 11	Heatmap	Correlation Matrix
Plot 12	Swarm	Engagement vs Device
Plot 13	Plotly Interactive	Likes vs Impressions

10. Analysis and Key Insights

10.1 Content Performance

- Reels and Videos generated the highest engagement rates.
- Tech and Fitness categories had the highest average likes.
- Brazil and Canada showed higher engagement rates.

10.2 User Trends

- Users aged 25–35 showed the highest engagement.
- Verified accounts had slightly higher engagement rates.
- Female users had marginally higher engagement than male users.

10.3 Behavioral Insights

- Evening hours generated the highest impressions.
- Mobile users had higher watch time than desktop users.
- Weekend posts showed better engagement trends.

10.4 Sentiment Analysis

- Positive sentiment posts outperformed other sentiment groups.
- Neutral posts had moderate engagement.
- Negative sentiment had the lowest engagement rates.

11. Insights & Conclusions

11.1 Descriptive Analysis

- Analyzed 5000 social media records containing user, post, engagement, device, sentiment, and time-related information.
- The analysis covered multiple countries, post types, categories, devices, and sentiment groups.
- The project combined data cleaning, transformation, statistical analysis, and visualization.

11.2 Diagnostic Analysis

- Content type and category differences were examined through engagement metrics.
- Age, gender, verification status, device type, and posting time were compared against engagement behavior.
- Sentiment was analyzed to understand its relationship with engagement performance.

11.3 Analytical Use

- Engagement metrics can help evaluate content performance.
- Time and device analysis can support decisions about posting schedules and platform usage.
- Sentiment analysis can help identify content associated with stronger or weaker engagement.

12. Python Analysis / Code Used

Important: This is a Python data-analysis project, not a SQL project. Therefore, there are no SQL queries in this documentation.

The analytical work was performed using Python libraries and techniques including Pandas DataFrames, NumPy statistical operations, data cleaning, missing-value imputation, duplicate removal, feature engineering, groupby analysis, descriptive statistics, and visualization with Matplotlib, Seaborn, and Plotly.

The completed Google Colab notebook contains the executable Python code for the individual assignment tasks and visualizations.

13. Project Deliverables

- Google Colab Python notebook containing the completed analysis.
- Social media engagement dataset used for the analysis.
- Project documentation in PDF format.
- GitHub README documentation.
- Screenshots showing the major analysis stages and visualizations.
- Final analytical insights.

14. Skills Demonstrated

- Data Loading and Type Conversion
- Missing Value Handling and Imputation
- Duplicate Detection and Removal
- Data Cleaning and Standardization
- Feature Engineering
- GroupBy Aggregations
- Descriptive and Statistical Analysis
- Matplotlib Visualization
- Seaborn Visualization
- Plotly Interactive Visualization
- Correlation and Distribution Analysis
- Insight Generation
- Python Data Analysis Workflow

15. Conclusion

The Social Media Engagement Analytics project demonstrates a complete Python-based data analysis workflow using a 5000-record social media dataset. The project covers data import, cleaning, missing-value handling, duplicate removal, feature engineering, exploratory analysis, statistical analysis, and visualization.

The analysis transforms raw social media engagement information into structured results that can be used to understand content performance, user behavior, posting-time patterns, device usage, and sentiment-related engagement trends.

16. Author

Field	Details
Name	Angel E
Course	Data Analytics (DA) – Module 5
Assignment	Python Module-End Assignment – Social Media Engagement Analytics
Tool	Google Colab
Language	Python 3
Topic	Social Media Engagement Analytics